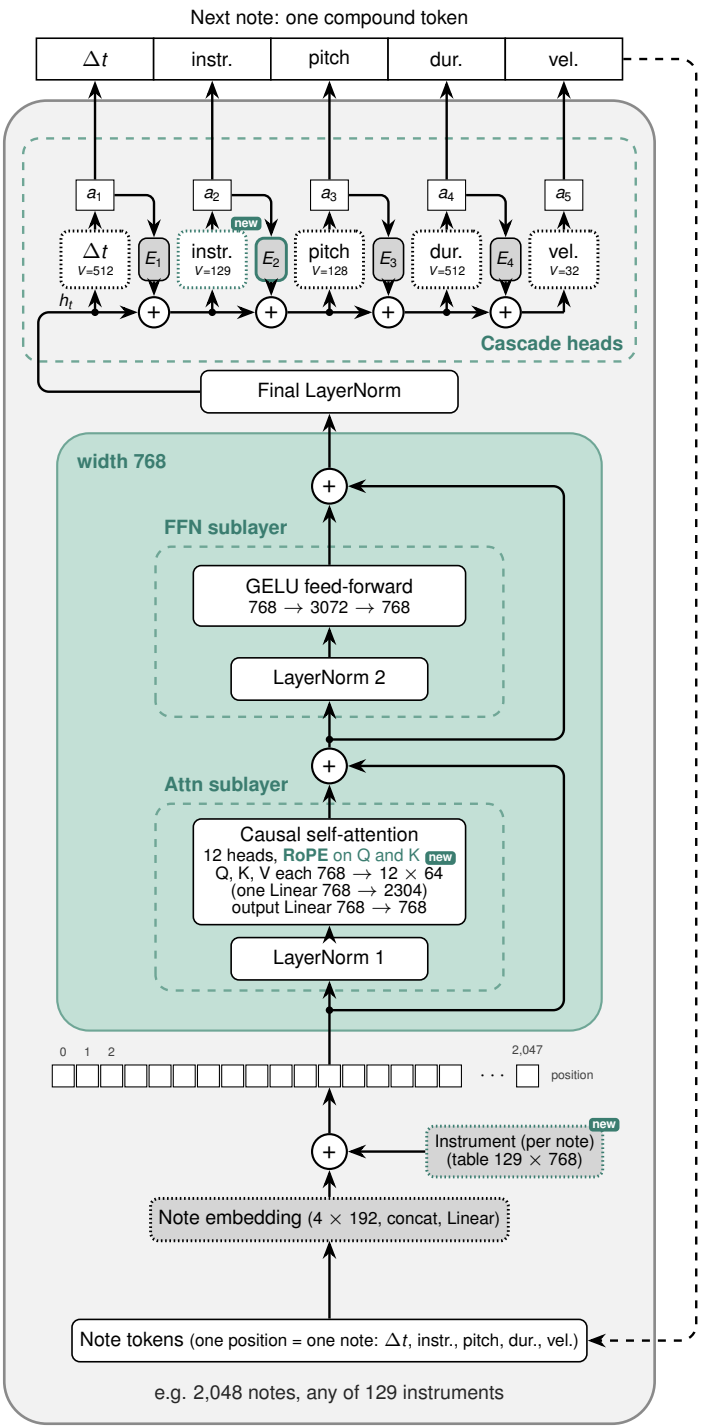
musicar-S (108M)

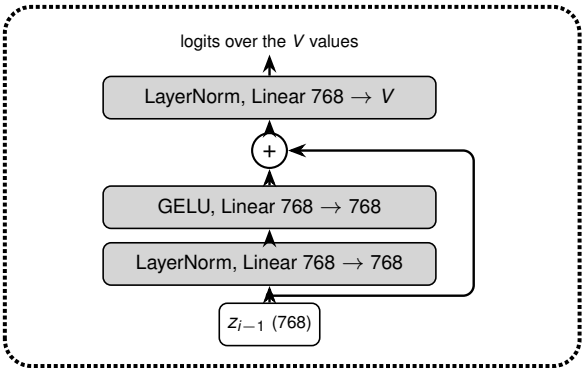
107.93M parameters: trunk 99.11M, cascade heads 7.90M, embeddings 0.92M. 129 instruments, context 2,048 notes.

How to read: numbers give the reading order; teal = the repeated block; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom (white panel) into the dotted part; new and teal outlines = added when the piano model was grown (section 4).

1 Music model



3 One cascade head



Why a cascade. Head i (a dotted box, own weights) reads the trunk state plus the embeddings E of the values already chosen,

$$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$$

so the note's probability follows the chain rule: $p(\Delta t)\, p(\text{instr.} \mid \Delta t) \cdots$. In training a_i is the true value (teacher forcing), in generation the sampled one. Loss: the sum of the five cross-entropies.

4 Grown from musicar-S-piano

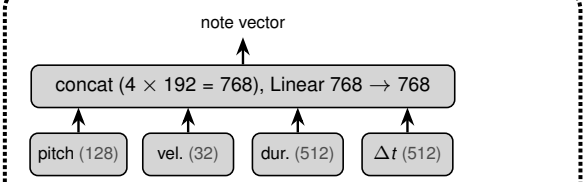
Copied (and trained on): the trunk, the note embedding, the four note heads and their condition embeddings.

New: the instrument embedding, the instrument head and its condition embedding E_2; both tables and the head's output layer start at zero: at step 0 they change nothing.

Changed: learned positions → RoPE, context 512 → 2,048 notes (not function-preserving).

Run: GigaMIDI 0.7 + Aria-MIDI 0.3, 197M notes (3,000 steps × 65,536), 83 min. GigaMIDI validation 2.232 nats / note; the same recipe from scratch: 2.597.

2 Input embedding (the dotted boxes)



+ instrument of this note new: a table of 129 vectors (128 General MIDI programs + drums), added to the note vector; the head *instr.* predicts the next one.

Positions: RoPE in every attention layer, no table.

Time is absolute: Δt and durations in 20 ms steps, clipped at 10.22 s. No start / end tokens, no window conditions.